

A scenic photograph of the UC Berkeley Campanile tower at sunset. The sun is low on the horizon, creating a warm orange glow and silhouetting the trees and buildings in the foreground. The tower stands prominently in the center, with its spire reaching towards the sky. The background shows a cityscape with various buildings and a distant mountain range under a cloudy sky.

Deep learning in Asset Pricing

Chen, Pelger & Zhu (2019)

Deep Learning and Applications

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Agenda



The paper



Replication



Extensions



Combination



Summary

The paper

The question: why do some stocks earn higher average returns than others, and what single pricing kernel (the SDF) explains all of them?



The factor zoo

Hundreds of firm characteristics claim to predict returns. Linear factor models use only a handful and miss how characteristics interact.



The economy keeps changing

Risk premia rise in recessions and fall in booms. A pricing model has to know the macro state, not only firm data.



Pure ML ignores economics

Return prediction with neural nets overfits a very noisy signal and has no no-arbitrage discipline.

THE PAPER'S ANSWER

Estimate the SDF with deep learning, disciplined by the no-arbitrage condition

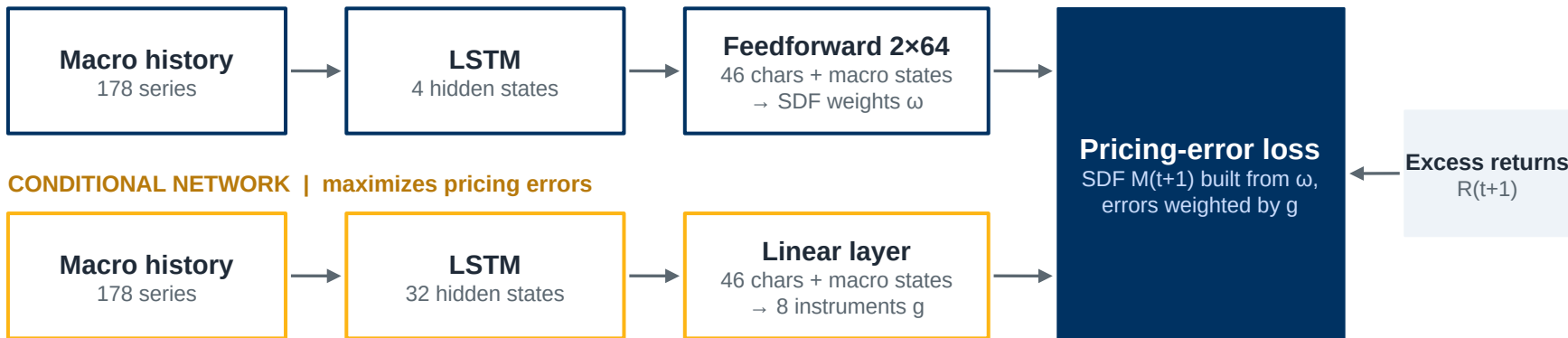
0.75 vs **0.19**

Out-of-sample monthly Sharpe: GAN vs Fama-French 3 (1992-2016)

A better SDF means better risk measurement, cost-of-capital estimates and portfolios. The paper shows economic theory and deep learning work better together than apart.

Model Architecture

SDF NETWORK | minimizes pricing errors



LSTM

Compresses 178 macro series into a few hidden states that track the economic cycle.

Feedforward network

Maps each stock's 46 characteristics and the macro states to its SDF weight, so characteristics can interact nonlinearly.

GAN

The conditional network searches for mispriced portfolios; the SDF network then updates to remove those pricing errors.

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Extensions



From the paper to our extensions

THE PAPER COMPARED

LS Linear SDF on 92 managed factors

EN Elastic Net, penalised linear SDF

FFN Feedforward return prediction

GAN Adversarial SDF with LSTM macro states

keep the
GAN, then



WHAT WE ADDED



Transformer

Replace the LSTM macro encoders with attention

MODEL



Rolling windows

Sharpe over every 36- and 60-month window

EVALUATION



Regime change

Sharpe in recessions vs expansions, high vs low volatility

EVALUATION



Uncertainty

Bootstrap confidence intervals for Sharpe gaps

EVALUATION

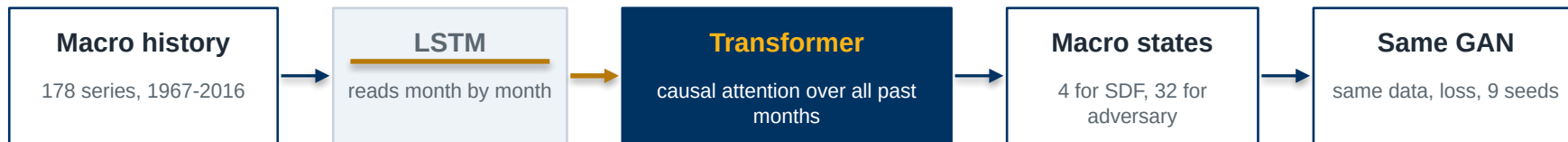
One model change, three new ways to judge it: is the paper's result robust to architecture, to time, to the economic state and to sampling noise?

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Extension 1: Transformer



Extension 1: Transformer instead of LSTM



Only the macro encoder changes. Both LSTM encoders in the architecture (SDF and conditional network) are replaced.

What it is

The architecture behind modern language models.
It looks at the whole macro history at once and learns which past months matter most for today (attention).

How we added it

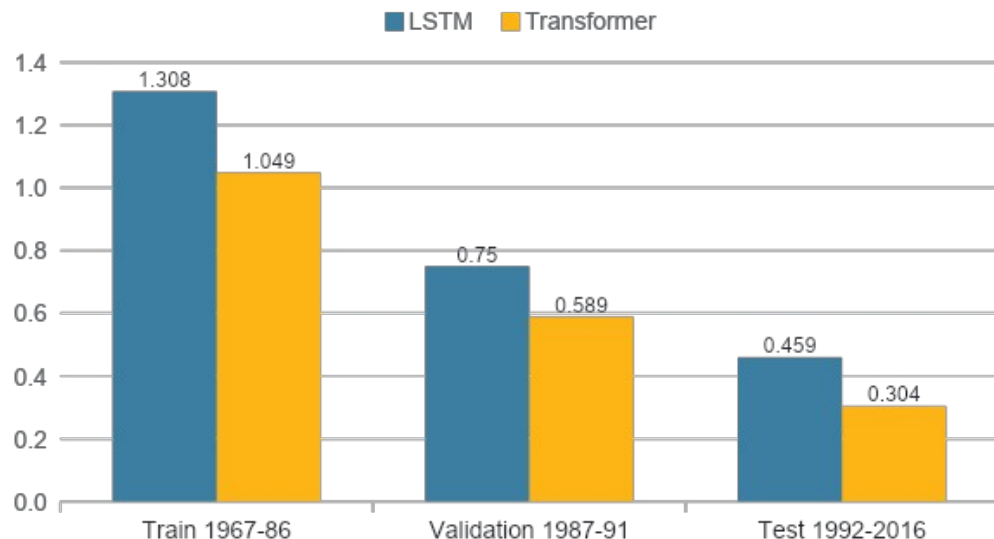
Swapped both LSTMs for causal Transformers (2 layers, 4 heads), the model cannot see the future.
Retrained the LSTM with the same budget and seeds, so the comparison is fair.

Why we added it

The LSTM squeezes decades into a small memory and can forget distant shocks.
Attention can link today directly to a past crisis. Does a newer model price assets better?

Question we ask: Can attention read the macro state better than the LSTM the paper uses?

Transformer: train, validation and test results



Monthly Sharpe ratio. Same data, loss, seeds and reduced training budget for both.



Extension LSTM leads Transformer: test monthly Sharpe is 0.459 versus 0.304.



This comparison holds under our reduced budget, not the full author schedule.

TEST PERIOD, 1992-2016

	LSTM	Transformer
Monthly Sharpe	0.459	0.304
Mean return / month	0.51%	0.42%
Annual volatility	3.9%	4.8%
Max drawdown	-11.1%	-15.4%

Lower return and higher risk. The Transformer loses on both parts of the Sharpe ratio, so the gap is not one bad stretch.

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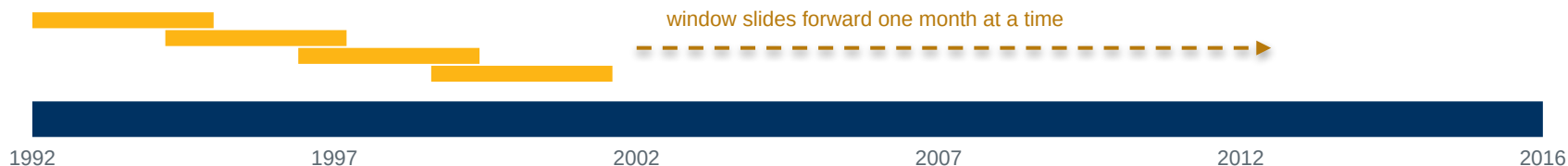
Extension 2: Rolling window



Extension 2: rolling-window evaluation

Paper: one Sharpe ratio for 1992-2016

Us: 265 Sharpe ratios, one per 36-month window



What it is

Sharpe ratio measured over every 3-year and every 5-year stretch of the test period, instead of one number for 25 years.

How we added it

Used each model's saved monthly returns; no retraining.

Slid the window one month at a time: 265 windows of 36 months and 241 of 60 months.

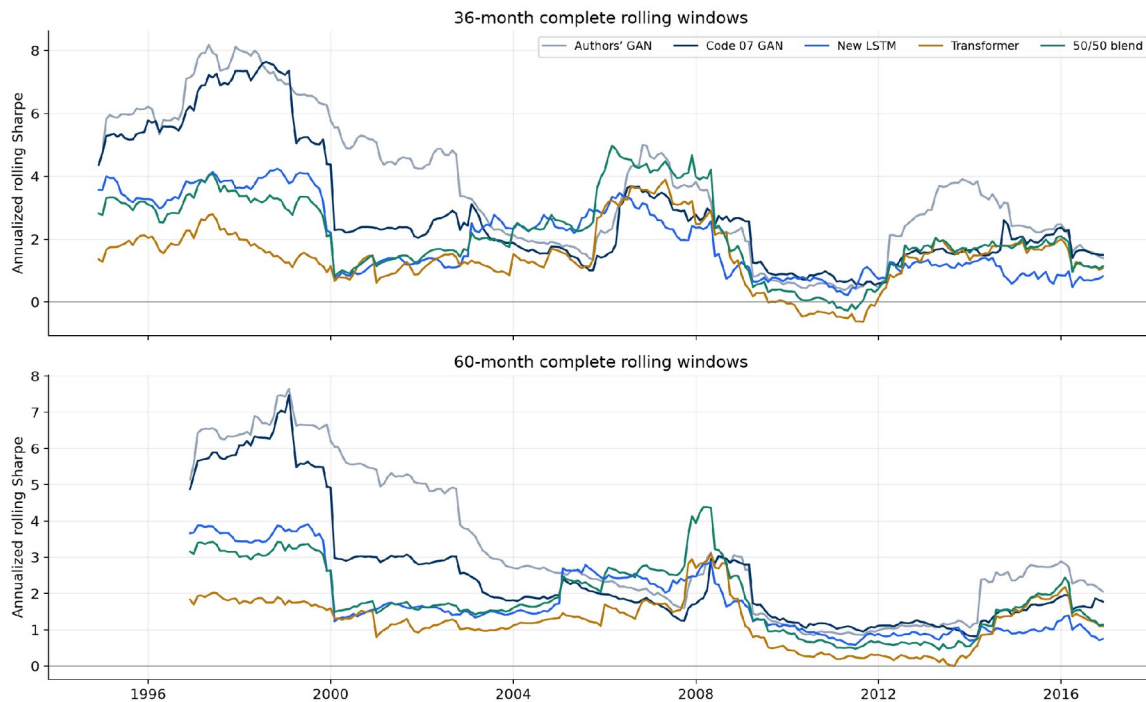
Why we added it

1992-2016 covers the dot-com bubble, the 2008 crisis and the recovery.

One average can hide years where a model fails, and investors live through those years.

Question we ask: Is the paper's 25-year Sharpe ratio stable, or driven by a few good years?

Rolling results: 36- and 60-month Sharpe

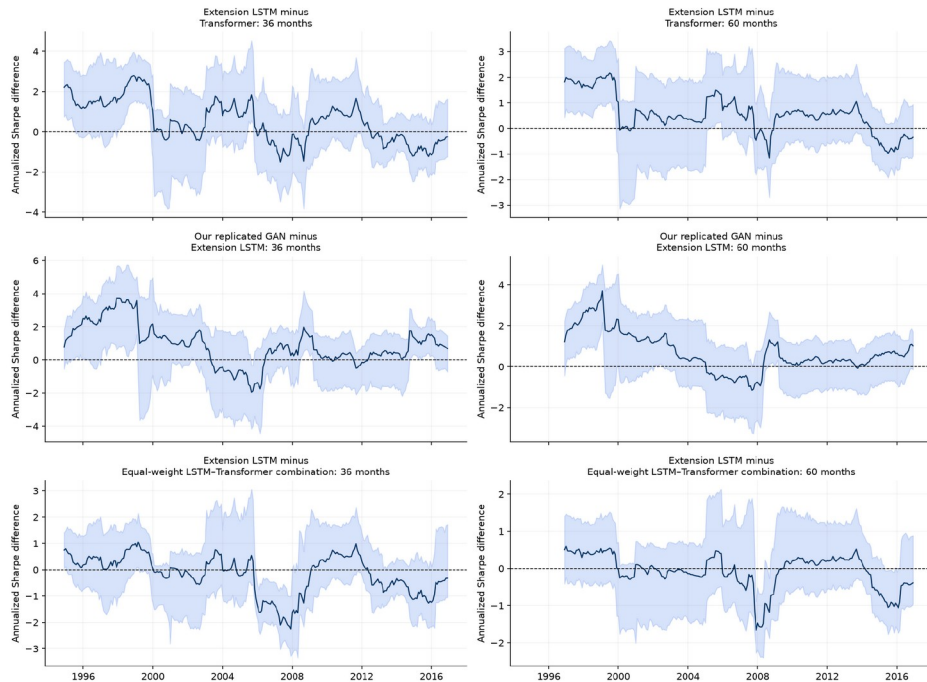


- ✓ Curves show annualized Sharpe over the previous 36 or 60 months.
- ✓ Longer windows smooth fluctuations. These fixed models are not retrained.

Mean rolling Sharpe	36m	60m
Paper GAN	3.64	3.20
Our GAN	2.89	2.50
LSTM	1.99	1.80
Transformer	1.47	1.23

Rolling: paired Sharpe differences

Paired rolling differences: pointwise 95% basic intervals



36- and 60-month windows

Line: observed annualized Sharpe gap.

Band: pointwise 95% basic interval.



Line: observed paired Sharpe gap. Shading: pointwise 95% basic interval.



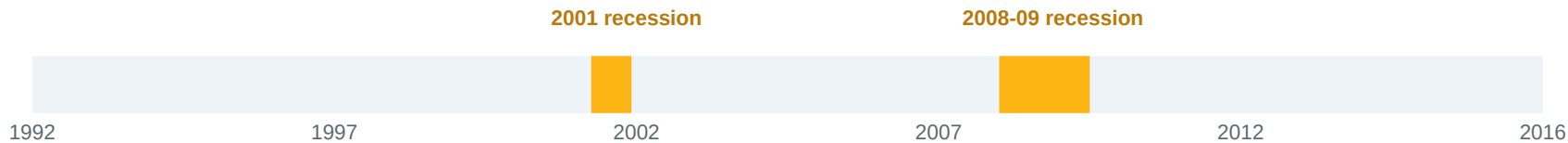
A band crossing zero leaves that window's ranking unresolved.

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Extension 3: Economic Regimes



Extension 3: economic regimes



Business cycle: 274 expansion vs 26 recession months (NBER) **Market volatility:** 123 high vs 177 low months

What it is

The 300 test months split by the state of the economy: expansion vs recession, and high vs low market volatility.

How we added it

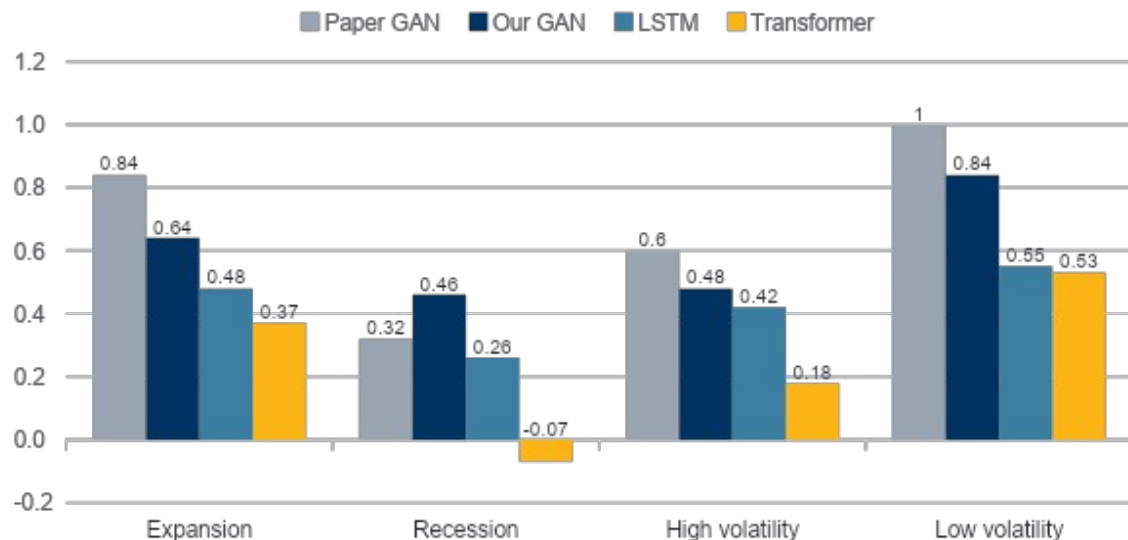
Same saved returns, the Sharpe computed inside each group.
Volatility uses the past 12 months only, with the cut-off set on training data, so no look-ahead.

Why we added it

The SDF should matter most in bad times. The macro encoder exists to read the economic state. If it works, it should show up here.

Question we ask: Does the model still price assets well when the economy turns bad?

Regime results: every model struggles in bad times



Monthly Sharpe ratio within each group of test months, 1992-2016.

-0.07

Transformer Sharpe in recessions, the only negative value

0.46 vs 0.32

Our GAN beats the authors' GAN in recessions, but this is only 26 months

All models: lower Sharpe in recessions and high-volatility months than in calm markets.

What it changes: the paper's result is mostly an expansion and calm-market result. Bad-time performance is weaker and rests on few months.

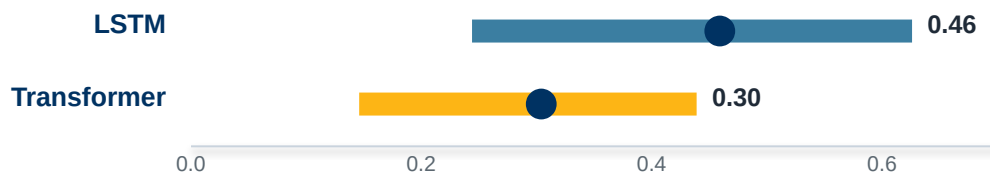
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Extension 4: Uncertainty



Extension 4: how sure are we? Bootstrap uncertainty

Same Sharpe ratios, now with a range of plausible values (illustration with our test results):



The ranges overlap. Is the gap real or luck?

What it is

A confidence interval around each Sharpe ratio, and around the gap between two models.

How we added it

Resampled 12-month blocks of the test period 10,000 times, every model gets the same blocks.

Repeated inside every rolling window.

Why we added it

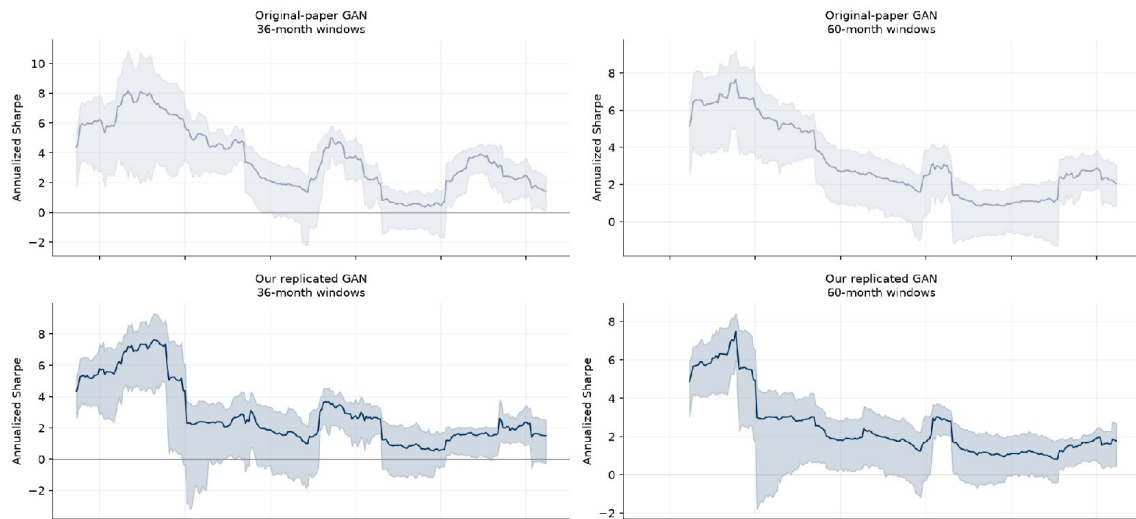
After the replication, GAN 0.61 vs FFN 0.57 looked like a ranking but could be noise.

Same question for LSTM vs Transformer, and for our gap to the paper.

Question we ask: Which differences between models are real, and which could be luck?

Uncertainty: paper and replication

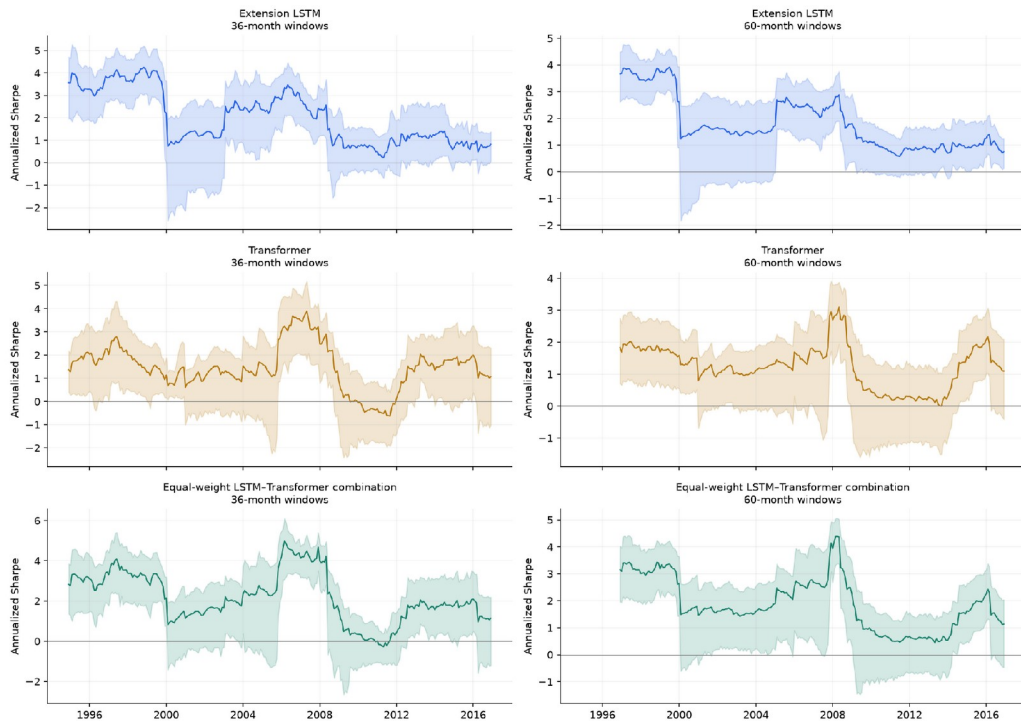
Rolling model Sharpe: pointwise 95% basic intervals, six-month blocks



✓ Lines show annualized rolling Sharpe for the paper and replicated GAN.

✓ Some bands cross zero, leaving positive Sharpe unresolved in those windows.

Uncertainty: extension models



✓ Separate confidence bands show sampling uncertainty for each extension model.

✓ Several bands cross zero. Use the paired comparison on slide 15 to assess model gaps.

Uncertainty: paired Sharpe differences

A minus B	Difference	95% basic	95% simultaneous
Paper GAN minus Replicated GAN	0.138	[-0.037, 0.256]	[-0.148, 0.425]
Extension LSTM minus Transformer	0.155	[-0.049, 0.365]	[-0.132, 0.441]
Extension LSTM minus Equal-weight blend	-0.004	[-0.132, 0.119]	[-0.290, 0.283]

A positive point estimate can coexist with an interval containing zero.

LSTM exceeds Transformer by 0.155 monthly Sharpe, but both intervals include zero.

The observed advantage remains statistically unresolved. This does not prove equality.

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Combined



All extensions together: one scorecard

Portfolio	Test Sharpe	95% interval	Rolling 36m	Expansion	Recession	High vol	Low vol
Paper GAN (authors)	0.75	0.44 to 0.97	3.64	0.84	0.32	0.60	1.00
Our GAN (replication)	0.61	0.38 to 0.78	2.89	0.64	0.46	0.48	0.84
LSTM (retrained)	0.46	0.24 to 0.63	1.99	0.48	0.26	0.42	0.55
Transformer	0.30	0.15 to 0.44	1.47	0.37	-0.07	0.18	0.53
50/50 blend	0.46	0.27 to 0.61	2.17	0.52	0.08	0.34	0.68

Monthly Sharpe except rolling (annualised mean over 265 windows). Gold = best of our own models in each column; grey row = authors' published factor.



Replication holds

Our GAN is our best model on every lens, and beats the paper in recessions.



Architecture

LSTM beats Transformer on every lens, but not by a statistically clear margin.



Time and state

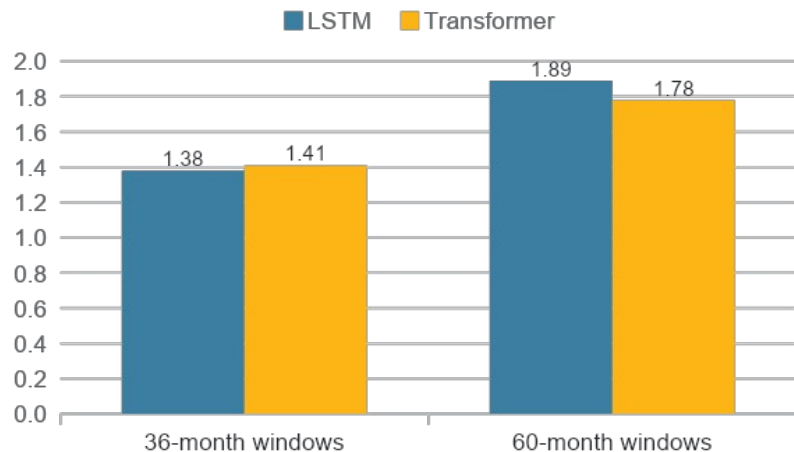
Every model is weaker after 2008, in recessions and in volatile markets.

Bottom line: the GAN-LSTM design survives every lens, but its edge is concentrated in expansions, calm markets and the 1990s.

Rolling Sharpe averages annualized ratios across overlapping 36-month windows; our replication's 2.89 equals 0.83 monthly

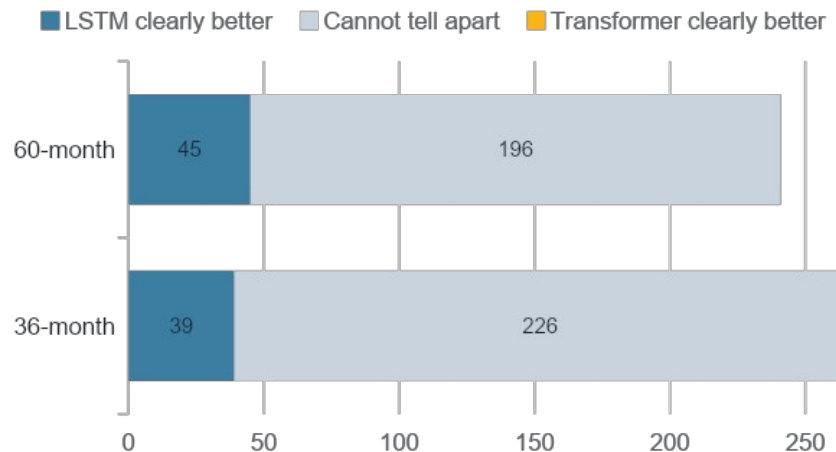
Combination: rolling results by regime and uncertainty

Rolling x regime: windows ending in a recession



Mean annualised rolling Sharpe. **Transformer leads on 3-year windows, LSTM on 5-year windows.**

Rolling x uncertainty: LSTM vs Transformer



Number of windows by 95% bootstrap verdict (265 and 241 windows).

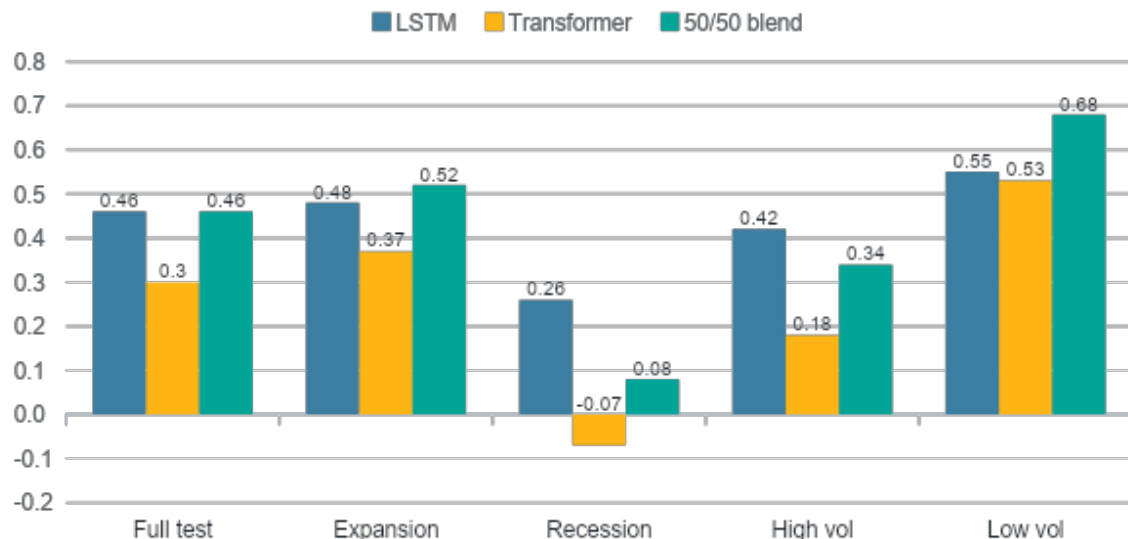
Combined message: architecture differences are small next to how much performance moves with time and how noisy Sharpe ratios are.

36-month windows: Blue represents 39 of 265 windows (14.7%) with statistical evidence favoring LSTM; gray represents 226 (85.3%) where the difference remains unresolved.

60-month windows: Blue represents 45 of 241 windows (18.7%) favoring LSTM; gray represents 196 (81.3%) where the difference remains unresolved.

Yellow: No windows statistically favor Transformer at either horizon. These conclusions use paired 95% bootstrap intervals, and overlapping windows are not independent tests.

Combining the two architectures: a 50/50 blend



Monthly Sharpe ratio, test period 1992-2016.

0.29

correlation between LSTM and Transformer returns

1.01%

blend monthly volatility, vs 1.12% LSTM and 1.39% Transformer

+0.004

Sharpe gain over LSTM; interval -0.13 to 0.12

Takeaway: diversifying across architectures lowers risk, but does not give a reliable Sharpe gain. Built after seeing test data: an illustration only.

Sharpe gain = Blend - LSTM = $0.463 - 0.459 = +0.004$; the 95% bootstrap interval $[-0.13, +0.12]$ includes zero, so the improvement is statistically unresolved.

Sample: 274 expansion months (91.3%) and 26 recession months (8.7%), based on NBER classifications.

Interpretation: Recession results rely on a small sample, so their Sharpe estimates require cautious interpretation.

Summary: findings and next steps

Our contribution combines architecture, rolling, regime and uncertainty evaluation.

Higher point estimates still need uncertainty checks and a new holdout.

What we add

- 1- We compared LSTM with **Transformer** under the same reduced training budget,
 - 2- examined **rolling** windows
 - 3- economic **regimes**,
 - 4- tested a **50/50 combination**,
 - 5- added bootstrap **uncertainty**.
- This evaluates performance beyond one full-period Sharpe ratio.

What the evidence supports

LSTM has higher observed test Sharpe than Transformer (**0.459 versus 0.304**).

However, the paired 95% bootstrap interval includes zero, so the advantage remains statistically unresolved.

The blend reduces volatility but adds only **0.004** to observed Sharpe.

What remains open

Full author-schedule GPU training would test whether additional training changes the results.
A fresh holdout would assess performance on unseen data.
Transaction costs would show whether trading reduces the gains, while analysis across training runs would measure sensitivity to initialization.

Appendix - Github Repo

<https://github.com/rezazamani2329/Deep-Learning-in-Asset-Pricing-Replication-and-Extensions>

